Topic 2.14 - Logarithmic Function Context and Data Modeling

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

Logarithmic models are appropriate when equal multiplicative changes in input produce approximately equal additive changes in output. They often describe diminishing returns or scale measurements, but their restricted domains and slow long-run growth must be interpreted in context.

Learning targets

- Construct logarithmic models from proportional input change, two points, transformations, or regression output.
- Use logarithmic models to predict outputs and solve for inputs.
- Interpret domains, units, scale meaning, diminishing change, and model limitations.

Language and structure

Term	Meaning in this topic
reference input	A convenient positive input at which the logarithm is 0, often $x = 1$ or $x = h + 1$.
diminishing returns	Output continues to increase while gains become smaller for equal additive input changes.
multiplicative predictor	An input whose meaningful comparisons are often ratios rather than differences.
logarithmic regression	A fitted model such as $y = a + b\ln x$ for positive inputs.
model domain	The positive or shifted-positive input interval on which the context and formula are meaningful.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A performance measure rises by about 6 points when practice time increases from 1 to 2 hours, from 2 to 4 hours, and from 4 to 8 hours. What model family is suggested, and why is a linear model less natural?

Concept 1: Build from multiplicative change

Core idea

If multiplying x by b changes output by a constant a, a model $f(x) = a\log_b x + k$ is natural.

- $f(bx) - f(x) = a$.
- The value at $x = 1$ is k because $\log_b 1 = 0$.
- Changing the base changes the coefficient but can represent the same function.

Worked example - annotate the reasoning

Problem. A score is 5 at input 1 and increases by 4 whenever the input doubles. Construct a model.
Model reasoning. $S(x) = 4\log_2 x + 5$ for $x > 0$.

Check your understanding

Independent

A response increases by 9 whenever the input is multiplied by 10, and the response is 22 at input 1. Write a model.

Concept 2: Build from two points and transformations

Core idea

For $f(x) = a\log_b(x-h)+k$, use the asymptote $x = h$ and points whose shifted inputs are convenient powers of b.

- The anchor point is $(h+1, k)$.
- Subtract k before comparing output changes.

- All modeled inputs must satisfy $x > h$.

Worked example - annotate the reasoning

Problem. Construct $f(x) = a \log_2 x + k$ through $(1, 5)$ and $(8, 17)$.
Model reasoning. At $x = 1$, $k = 5$. At $x = 8$, $\log_2 8 = 3$, so $3a + 5 = 17$ and $a = 4$. Thus $f(x) = 4 \log_2 x + 5$.

Check your understanding

Independent

A model has vertical asymptote $x = 3$, passes through $(4, 7)$, and increases by 6 when $x - 3$ is multiplied by 5. Find the model.

Concept 3: Prediction, inverse use, and limitations

Core idea

To solve for an input, isolate the logarithm and convert to exponential form.

- Predictions near the domain boundary can be unstable because the logarithm changes rapidly.
- Long-run logarithmic growth is slow but unbounded; physical limits may still invalidate extrapolation.
- A good fit does not imply that a logarithmic mechanism causes the relationship.

Worked example - annotate the reasoning

Problem. For $S(x) = 30 + 8 \ln x$, find the input that gives output 50.
Model reasoning. $8 \ln x = 20$, so $\ln x = 2.5$ and $x = e^{2.5} \approx 12.182$.

Check your understanding

Independent

Why is $S(x) = 30 + 8 \ln x$ not meaningful at $x = 0$?

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Construct a model with $f(1) = 12$ that increases by 5 when input is multiplied by 3.

Exit ticket 2

No calculator

Find the domain of $g(x) = 7 + 2\ln(x - 4)$.

Exit ticket 3

No calculator

Solve $10 + 3\log_2 x = 25$.